

Reviewer Recommendation and Comments for Manuscript Number CSEM-D-25-03193**Macroeconomic Uncertainty and Economic Value Creation: An Agent-Based Simulation of Firm Behavior Under Financial Shocks**

Original Submission
 Thanh Binh Lai, Ph.D candidate **Reviewer 2**

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Recommendation: **Reject****Overall Manuscript Rating (1 - 100):** 30**Transfer Authorization****Response**

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Yes

If this submission is transferred to another publication, do we have your consent to include your original review?

Yes

Reviewer Comments to Author

Thank you for the opportunity to read your manuscript. This research addresses an important topic and demonstrates substantial computational effort in constructing a large-scale agent-based macroeconomic model with financial frictions and policy interventions. The ambition of integrating firms, households, banks, and government within a unified simulation framework is appreciated. However, in its current form, the manuscript does not meet the standards required for publication in Computational Economics. The main concern is the lack of economic discipline and internal consistency in the model's core mechanisms.

First, despite frequent references to microfoundations, key agent behaviors—particularly firm production, pricing, and investment—are governed by heuristic adjustment rules and exogenous uncertainty factors rather than by well-defined objectives subject to constraints, making their economic interpretation unclear. This stands in contrast to both structurally founded macro models (e.g., Dynamic Discrete Choice Models (Rust) and Recursive Macroeconomic Theory (Ljungqvist and Sargent)) and to recent agent-based models that explicitly discipline behavioral rules using microeconomic logic (e.g., Agent-Based Models in Economics (Tesfatsion and Judd)). In the banking block, lending decisions rely on an expected-return calculation incorporating PD and LGD, which is a reasonable reduced-form approach. However, the manuscript does not clearly explain how these risk measures interact with regulatory constraints, nor which margin—expected losses, capital adequacy, or shifts in bank risk attitudes—primarily drives credit rationing during shocks. As a result, the dominant mechanism behind credit contraction and its aggregate effects remains difficult to interpret.

Second, the central concept of "economic value creation" is not clearly defined. At different points, it appears to refer to aggregate output, firm-level accounting notions, macroeconomic resilience, or innovation-driven growth. These concepts are not equivalent in economic theory (see, e.g., Capitalism, Socialism and Democracy; The Theory of Economic Growth), and the ambiguity is compounded by an internal inconsistency: the model assumes no capital accumulation or technological innovation, yet later sections interpret simulated "R&D investment" as a driver of GDP and recovery.

Third, household behavior and fiscal policy are treated in a highly reduced-form manner. Claims regarding precautionary saving and fiscal costs are not supported by explicit intertemporal household decisions or a government budget constraint, which limits the economic interpretability of the policy experiments (see, e.g., Fiscal Policy after the Financial Crisis).

Fourth, uncertainty is introduced as a uniform random shock that directly scales behavior in multiple, conceptually distinct blocks of the model. This ad hoc treatment does not correspond to standard economic notions of uncertainty and weakens the interpretation of the simulated dynamics.

Finally, the validation strategy relies primarily on qualitative matching of a large set of stylized facts. Given the model's complexity and number of free parameters, this approach does not convincingly identify the mechanisms emphasized in the paper.

Overall, while the paper is ambitious, the issues outlined above are fundamental and would require a substantial redesign of the model rather than incremental revision. For these reasons, I cannot recommend the manuscript for publication.

Reviewer Confidential Comments to Editor:**1. Is the paper acceptable for publication?**

The paper is not acceptable for publication in its present form, nor would minor or major revisions be sufficient to bring it to the standard expected for Computational Economics. I therefore recommend Reject.

2. Rationale for Recommendation

The manuscript's central limitations are structural rather than incremental.

- The model relies heavily on heuristic behavioral rules and exogenous uncertainty terms while repeatedly invoking microfoundations and economic interpretation. Firm production, pricing, and investment decisions are not disciplined by well-defined objectives subject to constraints, which makes the economic mechanisms difficult to interpret. Although reduced-form measures such as PD and LGD are incorporated into the banking block, the manuscript does not clearly establish which economic margins—expected losses, regulatory constraints, or shifts in bank risk attitudes—drive credit rationing during shocks.

- In addition, the central concept of "economic value creation" is used inconsistently, alternating between aggregate output, firm-level accounting notions, macroeconomic resilience, and innovation-driven growth. This ambiguity is compounded by an internal inconsistency: the model assumes no capital accumulation or technological innovation, yet later sections interpret simulated "R&D investment" as a driver of GDP and recovery.

- Further concerns include highly reduced-form household behavior and fiscal policy without an explicit intertemporal framework or government budget constraint, an ad hoc treatment of uncertainty as a uniform shock affecting multiple unrelated blocks, and a validation strategy based

primarily on broad stylized-fact matching despite substantial model flexibility.

Given these structural issues, I do not believe the manuscript could reach the journal's standards through revision.

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